Qubits

The fundamental unit of quantum information is the quantum bit, or **qubit**.

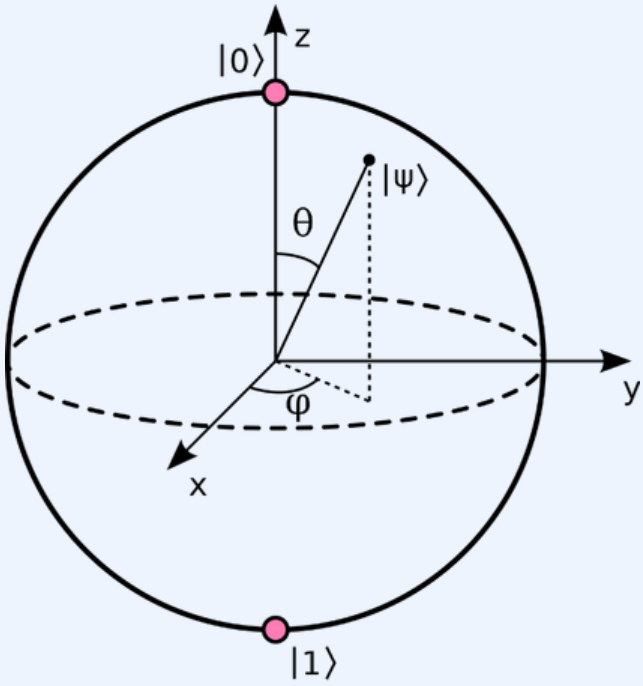
A qubit can be in the states, $|0\rangle, |1\rangle$ or a superposition of $|0\rangle$ and $|1\rangle$.

Superposition: a qubit can be in a **linear combination** of multiple states at the same time until it has been measured

General single-qubit state: $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$

c_0, c_1 are the **amplitudes** of $|0\rangle$ and $|1\rangle$, they are complex numbers satisfying $|c_0|^2 + |c_1|^2 = 1$.

This is the **Bloch sphere**. We can use it to visualise single-qubit states. All single-qubit states lie on the suface of the Bloch sphere.



Some single-qubit states are common so it's useful to have a shorthand to refer to them. These are the 'plus' and 'minus' states.

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Measurement

To get any information out of a qubit we need to measure it.

Measuring collapses the superposition, this means that a qubit that was once in a linear combination of two states gives a **definite outcome** as the output. The outcome is determined **probabilistically** depending on the amplitudes of each state.

Born rule: The probability of a measurement returning a state with a given amplitude is equal to **the magnitude of the amplitude squared.**

Example 1: measure $|+\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
$$= \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

State	Amplitude	Probability
$\|0\rangle$	$\frac{1}{\sqrt{2}}$	$\left\|\frac{1}{\sqrt{2}}\right\|^2 = \frac{1}{2}$
$\|1\rangle$	$\frac{1}{\sqrt{2}}$	$\left\|\frac{1}{\sqrt{2}}\right\|^2 = \frac{1}{2}$

So we have a 50% chance of returning the output $|0\rangle$ or $|1\rangle$.

(NOT REQUIRED for the labs but useful conceptually:
A common misconception in quantum computing is that the power comes from “quantum parallelism,” i.e. the ability to input a superposition of multiple starting states and compute across every state in parallel, then measure the right answer. But measurement gives an answer at random, we can’t just pick out the answer we want. Instead, quantum algorithms often use interference between states to increase the chance of measuring the correct output.)

CHECK YOUR UNDERSTANDING

Question 1: Find the eigenvectors and eigenvalues of $\begin{pmatrix} 9 & 8 \\ -6 & -5 \end{pmatrix}$

Question 2: convert 49 into binary

Question 3: convert 1000011 into base-10

Question 4: measure the state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

Question 5: for the single-qubit state $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$ why must $|c_0|^2 + |c_1|^2 = 1$?

Phase

In example 1 and question 3 we noticed that:
Measuring $|+\rangle$ gives outcomes $|0\rangle, |1\rangle$ each with 50% probability
Measuring $|-\rangle$ gives outcomes $|0\rangle, |1\rangle$ each with 50% probability

So what's the difference between these two states?

They have a **phase difference:**

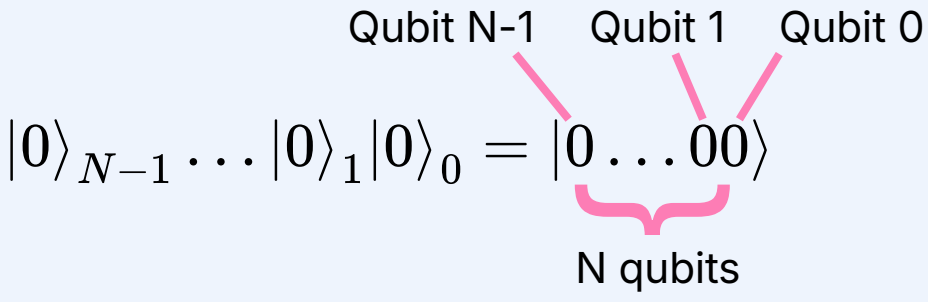
$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

This can affect how the state **changes over time**. We will study how states can change over the next few pages as we look at quantum gates.

Multiple qubit states

N bits can be in **one** of 2^N possible states.
N qubits can be in a **superposition** of all 2^N possible states.

When writing code in Qiskit, multiple-qubit states are written using little-endian notation, where we order qubits from right to left rather than left to right. This is useful in code because it mirrors the order of bits in binary numbers.



WARNING: Be careful when specifying which qubit you are working with in your code! In QIT/Quantum Computation qubits are ordered from left to right, any differences between course notes and our notes is due to this notation.

Quantum gates

Quantum gates can be combined to create quantum circuits!
Quantum gates must be reversible → represented by unitary matrices

Single qubit gates

Name	Action on computational basis	Representative matrix
X gate	$X\|0\rangle = \|1\rangle$ $X\|1\rangle = \|0\rangle$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Hadamard (H) gate	$H\|0\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle) = \|+\rangle$ $H\|1\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle) = \|-\rangle$	$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$
Z gate	$Z\|0\rangle = \|0\rangle$ $Z\|1\rangle = -\|1\rangle$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
T gate	$T\|0\rangle = \|0\rangle$ $T\|1\rangle = e^{i\pi/4}\|1\rangle$	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

Example 1: apply an X gate to $|\psi_1\rangle = \alpha|0\rangle + \beta|1\rangle$

$$
\begin{aligned}
X|\psi_1\rangle &= X\left(\alpha|0\rangle + \beta|1\rangle\right) \\
&= X\left(\alpha|0\rangle\right) + X\left(\beta|1\rangle\right) \\
&= \alpha X|0\rangle + \beta X|1\rangle \\
&= \alpha|1\rangle + \beta|0\rangle
\end{aligned}
$$

Double qubit gates

Name	Action on computational basis	Representative matrix
SWAP	$SWAP\|00\rangle = \|00\rangle$ $SWAP\|01\rangle = \|10\rangle$ $SWAP\|10\rangle = \|01\rangle$ $SWAP\|11\rangle = \|11\rangle$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$
Controlled-NOT (CNOT / CX)	$CX\|00\rangle = \|00\rangle$ $CX\|01\rangle = \|11\rangle$ $CX\|10\rangle = \|10\rangle$ $CX\|11\rangle = \|01\rangle$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$

Example 2: apply a CX gate to $|\psi_2\rangle = \alpha|0\rangle_1|0\rangle_0 + \beta|0\rangle_1|1\rangle_0$ where qubit 0 is the control qubit and qubit 1 is the target qubit

$$\begin{aligned} CX|\psi_2\rangle &= CX\left(\alpha|0\rangle_1|0\rangle_0 + \beta|0\rangle_1|1\rangle_0\right) \\ &= CX\left(\alpha|0\rangle_1|0\rangle_0\right) + CX\left(\beta|0\rangle_1|1\rangle_0\right) \\ &= \alpha CX|0\rangle_1|0\rangle_0 + \beta CX|0\rangle_1|1\rangle_0 \\ &= \alpha|0\rangle_1|0\rangle_0 + \beta|1\rangle_1|1\rangle_0 \end{aligned}$$

Triple qubit gates

Name	Action on computational basis	Representative matrix
Toffoli gate (CCX): There are two control qubits and one target qubit. If both control qubits are in the state , the state of the target qubit is flipped.	The computational basis in this case consists of 16 qubits $CCX\|000\rangle = \|000\rangle$ $CCX\|001\rangle = \|001\rangle$ $CCX\|010\rangle = \|010\rangle$ $CCX\|011\rangle = \|111\rangle$ $CCX\|100\rangle = \|100\rangle$ $CCX\|101\rangle = \|101\rangle$ $CCX\|110\rangle = \|110\rangle$ $CCX\|111\rangle = \|011\rangle$	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$

REMINDER: we are using little-endian notation, so the matrix for the CX, SWAP and CCX is different to that in the QIT/Quantum Computing notes.

CHECK YOUR UNDERSTANDING

Question 1: What is the outcome of applying a H gate to the state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$?

Question 2: What is the outcome of applying a H gate to the state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$?

Question 3: Show that the CX gate is reversible.

Question 4: The AND gate takes 2 bits as input, and outputs 1 if and only if both bits = 1. Which quantum gate is analogous to the AND gate?